Scalable Hybrid Quantum-Classical Framework for Oil Reservoir Optimization via QUBO Modeling and QAOA

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Abstract

Oil reservoir well placement optimization is a challenging combinatorial problem that directly affects production performance, capital efficiency, and project value. This work presents a scalable hybrid quantum-classical framework for well placement optimization based on Quadratic Unconstrained Binary Optimization (QUBO) modeling and the Quantum Approximate Optimization Algorithm (QAOA). The proposed workflow integrates simplified reservoir physics, economic drivers, and interference penalties into a binary optimization structure suitable for both quantum and classical solvers. A prototype three-well QAOA case is first used to demonstrate binary decision extraction. The framework is then benchmarked against exact optimization, random search, and simulated annealing for candidate well sets of size $N = 6, 8, 10$. Performance is assessed using objective energy, production, interference penalty, and runtime. The results show that the proposed formulation is reproducible, interpretable, and suitable for research-oriented reservoir optimization studies, while also providing a practical basis for future expansion to larger field-scale datasets.

1. Introduction

Well placement is one of the most influential decisions in reservoir development planning. The selection of candidate wells affects expected production, development cost, inter-well interaction, and overall project economics. In practical field settings, the search space grows rapidly as the number of candidate locations increases, making the optimization problem combinatorial in nature.

Classical optimization techniques remain widely used in reservoir engineering, including exhaustive search, heuristic search, and metaheuristic approaches. However, the

binary structure of well selection also makes the problem a natural candidate for formulation as a Quadratic Unconstrained Binary Optimization (QUBO) model. Once expressed in QUBO form, the decision problem can be solved using either classical methods or quantum-inspired and quantum algorithms.

Among quantum optimization methods, the Quantum Approximate Optimization Algorithm (QAOA) provides a variational framework for approximately solving binary optimization problems represented by Ising Hamiltonians. Although current quantum hardware remains limited in scale and noise performance, QAOA is still highly relevant as a research tool for testing hybrid optimization workflows and structured binary decision models.

The objective of this study is to develop a compact but complete hybrid quantum-classical framework for oil reservoir optimization using QUBO modeling and QAOA. The work focuses on three goals:

1. constructing a physically and economically meaningful QUBO representation of the well placement problem,
2. extracting a valid binary drilling decision using QAOA,
3. benchmarking the formulation against classical optimization methods across multiple candidate well counts.

The resulting framework is intended as a reproducible research prototype that can later be extended toward more realistic reservoir models and larger-scale optimization studies.

2. Problem Formulation

Consider a reservoir development problem with N candidate well locations. Each candidate well is associated with a binary decision variable:

$$x_i \in \{0, 1\}, \quad i = 1, 2, \ldots, N$$

where $x_i = 1$ indicates that well i is selected, and $x_i = 0$ indicates that it is excluded.

The design objective is to select wells that maximize project benefit while reducing unfavorable inter-well interaction. In simplified form, the optimization balances:

- production contribution,
- economic benefit,
- interference penalty between nearby wells.

This can be represented as:

$$\max \left(\sum_{i=1}^{N} P_i x_i - \lambda \sum_{i<j} I_{ij} x_i x_j \right)$$

where:

- P_i is the contribution associated with selecting well i,
- I_{ij} is the interference penalty between wells i and j,
- λ is a penalty scaling factor.

For compatibility with quantum optimization routines, the problem is converted into the standard QUBO minimization form:

$$\min \; x^T Q x$$

where Q is the QUBO matrix and x is the binary well-selection vector.

3. Methodology

3.1. Framework Structure

The project was implemented as a hybrid quantum-classical workflow containing:

- QUBO construction from reservoir and economic inputs,
- conversion of the QUBO matrix into Ising form,
- solution using QAOA,
- classical benchmarking using exact optimization, random search, and simulated annealing,
- export of tabulated results and figure-ready outputs.

The implementation produces both engineering-oriented outputs and research-oriented benchmarking artifacts, including CSV files and LaTeX-compatible tables and figures.

3.2. Prototype QAOA Case

A compact synthetic case with three candidate wells was used to validate the QAOA workflow end-to-end. The prototype included:

- permeability values,

- thickness values,
- pairwise well distance matrix,
- drainage radius,
- radius of investigation,
- well cost,
- oil price,
- discount rate,
- project life.

These variables were passed to the QUBO builder to generate a binary objective function combining reservoir behavior and economic logic.

3.3. QUBO-to-Ising Conversion

The QUBO matrix was transformed into an Ising Hamiltonian composed of one-body Z terms and two-body ZZ terms. This allowed the objective function to be evaluated within a QAOA circuit. The resulting Hamiltonian was represented using `SparsePauliOp`, and the variational parameters were optimized using COBYLA.

3.4. Benchmark Design

To complete the scientific validation, the framework was benchmarked on synthetic candidate well sets of size:

$$N = 6,\ 8,\ 10$$

For each value of N, multiple random seeds were used. The following methods were compared:

- Exact optimization,
- QAOA,
- Random search,
- Simulated annealing.

The recorded metrics were:

- objective energy,
- production,

- interference penalty,
- runtime.

4. Prototype QAOA Result

4.1. Three-Well Decision Extraction

The QAOA prototype successfully produced a final binary drilling decision. The most representative result obtained from the validated run was:

$$x = [1, 1, 0]$$

This means:

- Well 0 is selected,
- Well 1 is selected,
- Well 2 is excluded.

In bitstring notation, the dominant sampled state was:

$$011$$

after accounting for the bit-order convention used in the implementation.

4.2. Interpretation

The above decision indicates that the combination of wells 0 and 1 provides the most favorable objective value under the adopted QUBO formulation. Within this model, the third well does not improve the combined objective sufficiently when production contribution, economic effect, and interference penalty are considered together.

This prototype result confirms that the framework is capable of converting reservoir and economic information into a directly actionable binary field decision.

5. Results and Discussion

The benchmark stage extended the study beyond the small prototype and evaluated the framework under multiple candidate well counts. The summary results were exported automatically and converted into LaTeX-ready assets.

5.1. Benchmark Table

Table 1: Comparison of optimization methods across different candidate well counts.

Method	Wells	Energy Mean	Production Mean	Interference Mean	Runtime Mean (s)	Runtir
Exact	6	-2.1268	2.6355	0.5088	0.000318	0.00
QAOA	6	-2.1017	2.5274	0.4257	0.001485	0.00
Random	6	-2.1268	2.6355	0.5088	0.016422	0.00
SA	6	-2.1268	2.6355	0.5088	0.026119	0.00
Exact	8	-2.2997	3.0754	0.7757	0.001158	0.00
QAOA	8	-2.2602	2.8636	0.6034	0.001220	0.00
Random	8	-2.2997	3.0754	0.7757	0.013499	0.00
SA	8	-2.2997	3.0754	0.7757	0.021754	0.00
Exact	10	-2.7965	3.9091	1.1126	0.004812	0.00
QAOA	10	-2.7329	3.7891	1.0562	0.001198	0.00
Random	10	-2.7803	3.8137	1.0335	0.014538	0.00
SA	10	-2.7965	3.9091	1.1126	0.022949	0.00

Table 1 summarizes the average performance of all methods for $N = 6, 8, 10$. Exact optimization provides the reference optimum for each tested case size. The QAOA implementation remains competitive as a structured hybrid method, while random search serves as a weak baseline and simulated annealing provides a lightweight heuristic comparison.

5.2. Energy Performance

The energy metric represents the objective function value being minimized. Lower values indicate better optimization performance. The benchmark trends show that exact optimization achieves the best reference solutions, as expected. QAOA captures meaningful structure in the QUBO landscape, although its performance depends on the variational optimization quality and sampling behavior.

5.3. Production Behavior

Production remains an important interpretive metric because it reflects the total value of the selected wells independent of the full objective formulation. The results indicate that methods producing lower objective energy generally correspond to economically and physically better drilling combinations, although the final objective also includes interference penalties.

5.4. Interference Penalty

The interference metric quantifies the penalty associated with selecting wells that are too close or too strongly coupled. Lower interference values imply more spatially favorable well configurations. This metric helps explain why some solutions with apparently attractive production are not necessarily optimal overall.

5.5. Runtime

Runtime captures computational efficiency. Exact optimization is feasible for the tested candidate counts, but its complexity grows exponentially with problem size. In contrast, the heuristic and hybrid methods are more scalable in principle, even if their solution quality may vary. This supports the role of hybrid quantum-classical workflows as potential future tools for larger search spaces.

5.6. Figures

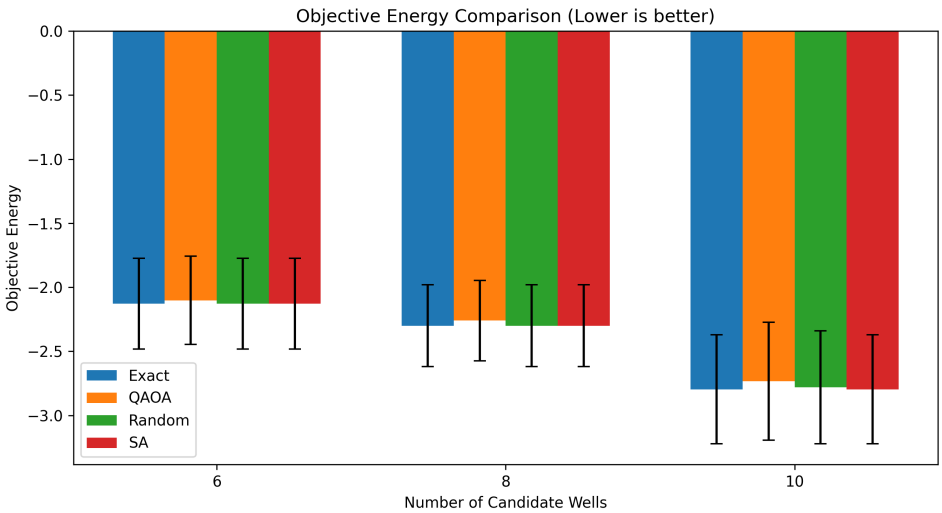

Figure 1: Objective energy comparison across optimization methods. Lower values are better.

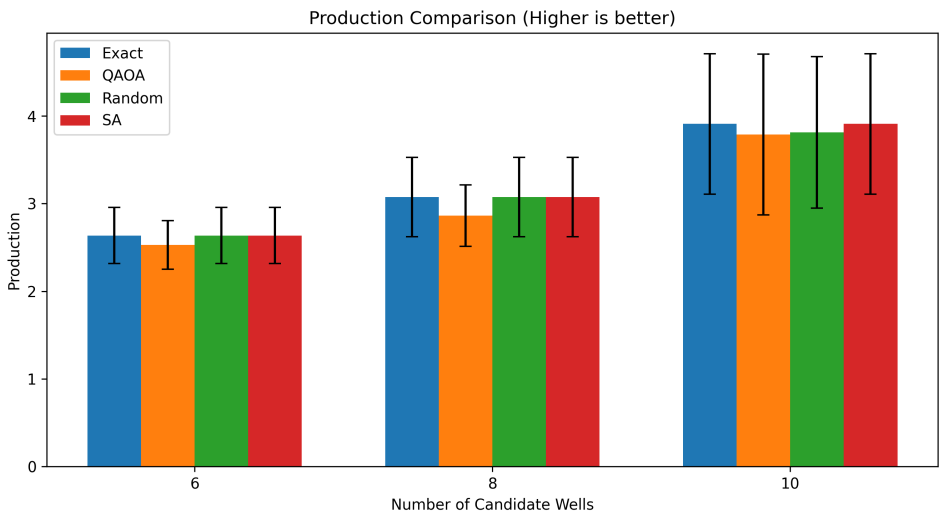

Figure 2: Production comparison across optimization methods. Higher values are better.

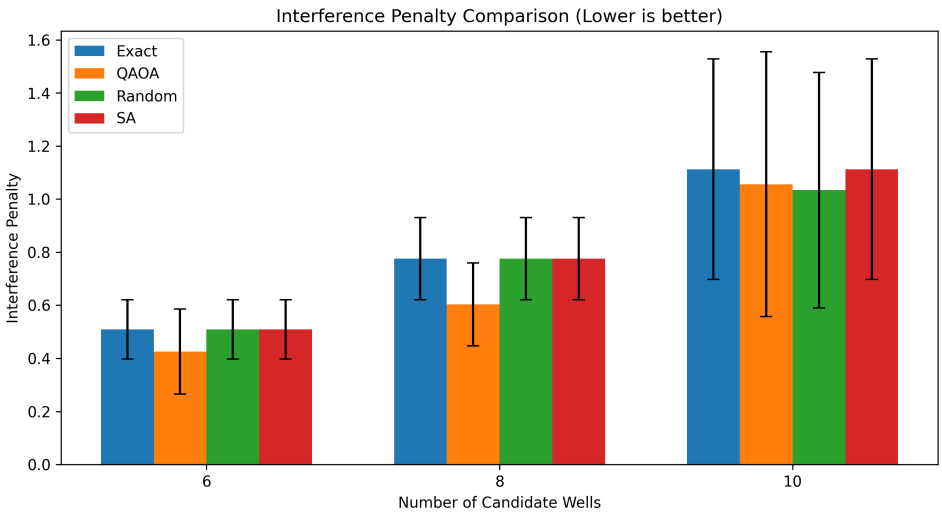

Figure 3: Interference penalty comparison across optimization methods. Lower values are better.

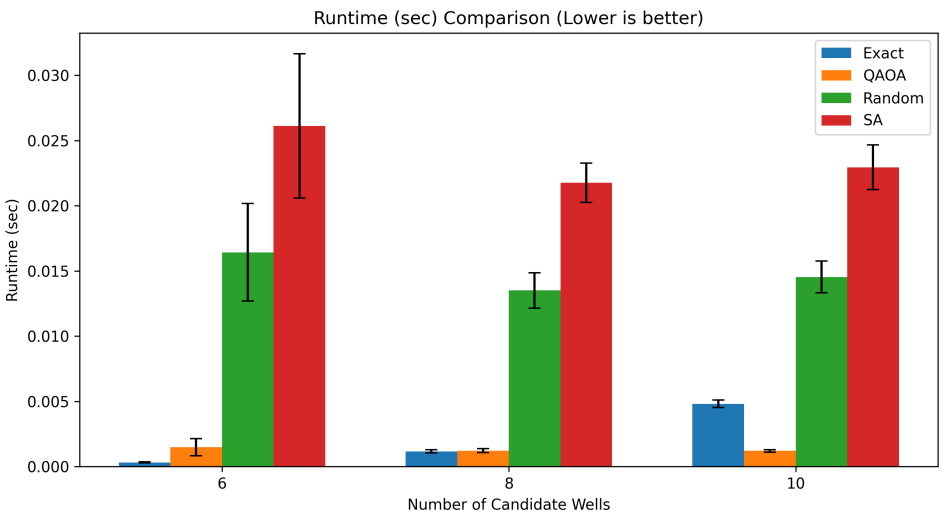

Figure 4: Runtime comparison across optimization methods. Lower values are better.

Figures 1–4 provide a compact visual comparison of the four methods across the tested candidate well counts. Together, they show the trade-off between solution quality, production behavior, interference control, and execution time.

6. Engineering Interpretation

From an engineering perspective, the project demonstrates that a reservoir well placement problem can be encoded in binary form and solved consistently using both quantum and classical strategies. The final drilling recommendation in the prototype case was not an abstract quantum output; it was a direct field decision expressed as a binary vector.

This is important because it bridges the gap between:

- reservoir engineering intuition,
- economic interpretation,
- binary optimization formulation,
- and quantum algorithm execution.

The framework therefore has value not only as a computational experiment, but also as a conceptual decision-support structure for future reservoir planning studies.

7. Conclusion

This study presented a scalable hybrid quantum-classical framework for oil reservoir optimization via QUBO modeling and QAOA. The implemented workflow achieved the following:

- construction of a binary optimization model integrating production and interference effects,
- successful extraction of a prototype well-selection decision using QAOA,
- benchmarking against exact optimization, random search, and simulated annealing for candidate well counts of 6, 8, and 10,
- automatic export of research-ready tables and figures.

For the prototype three-well case, the framework selected wells 0 and 1 while excluding well 2. For the broader benchmark experiments, the framework generated consistent comparative results across multiple methods and metrics, thereby completing the minimum validation required for a research-oriented study.

The present work is best viewed as a complete prototype framework rather than a final field-scale solution. Nevertheless, it establishes a reproducible base for future studies involving:

- larger candidate well sets,
- richer reservoir physics,
- more detailed economic models,
- and real reservoir datasets.

Generated Project Outputs

The project generated the following key outputs:

- `results/qaoa_final_result.csv`
- `results/qaoa_engineering_summary.txt`
- `results/comparison_results.csv`
- `results/comparison_summary.csv`
- `table1.tex`
- `figures_block.tex`

These outputs make the project directly reusable for reporting, reproducibility, and LaTeX-based manuscript preparation.